

Number 1

Code:



```
1 # number 1
2 # data
3 X <- c(6, 5, 8, 8, 7, 6, 10, 4, 9, 7)
4 Y <- c(8, 7, 7, 10, 5, 8, 10, 6, 8, 6)
5
6 n <- length(X)
7
8 # Regression of Y on X
9 model_yx <- lm(Y ~ X)
10 SSE_yx <- sum(residuals(model_yx)^2)
11 s_yx <- sqrt(SSE_yx / (n - 2))
12
13 # Regression of X on Y
14 model_xy <- lm(X ~ Y)
15 SSE_xy <- sum(residuals(model_xy)^2)
16 s_xy <- sqrt(SSE_xy / (n - 2))
17
18 # Results
19 s_yx
20 s_xy
```

Output:

Output

```
[1] 1.457738
[1] 1.613084
```

Number 2 letter a and b

Code:

```
22 # number 2 letter a
23 data(pressure)
24
25 X <- pressure$temperature
26 Y <- pressure$pressure
27
28 cor_xy <- cor(X, Y)
29 cor_xy
30
31 # number 2 letter b
32 X_prime <- 2*X + 6
33 Y_prime <- 3*Y - 15
34
35 cor_transformed <- cor(X_prime, Y_prime)
36 cor_transformed
--
```

Output:

```
[1] 0.7577923
[1] 0.7577923
```

The result in letter a and b are the same because the correlation does not change under linear transformations.

Number 3

For Letter a

Code and output:

```
> library(nycflights13)
>
> data(flights)
>
> # Remove missing values
> df <- na.omit(flights[, c("dep_delay", "arr_delay")])
>
> # Compute correlation
> cor_xy <- cor(df$dep_delay, df$arr_delay)
> cor_xy
[1] 0.9148028
```

Letter b

Code and output:

```
> X <- df$dep_delay
> Y <- df$arr_delay
>
> X_prime <- 2*X + 10
> Y_prime <- 0.5*Y - 5
>
> cor_transformed <- cor(X_prime, Y_prime)
> cor_transformed
[1] 0.9148028
```

Letter c

- The result in part a and b are just the same, because correlation is invariant under linear transformations meaning adding or multiplying a constant has no effect.

Letter d

- The correlation is strong and positive, meaning if it departs late, it is very likely that the arrival will be delayed too.

Number 4

For Letter a code and output:

```
> library(nycflights13)
>
> data(flights)
>
> # Remove missing values
> flights_clean <- na.omit(flights)
>
> # Create variables
> flights_clean$X1 <- ifelse(flights_clean$dep_delay <= 0, 0, 1)
> flights_clean$X2 <- flights_clean$dep_delay
> flights_clean$X3 <- flights_clean$air_time
> flights_clean$X4 <- flights_clean$distance
> flights_clean$X5 <- flights_clean$hour
> flights_clean$X6 <- ifelse(flights_clean$carrier %in% c("AA","DL","UA","WN"), 1, 0)
>
> # Fit regression
> model <- lm(arr_delay ~ X1 + X2 + X3 + X4 + X5 + X6, data = flights_clean)
>
> summary(model)
```

```
Call:
lm(formula = arr_delay ~ X1 + X2 + X3 + X4 + X5 + X6, data = flights_clean)
```

```
Residuals:
    Min       1Q   Median       3Q      Max
-111.400   -9.491   -1.816    7.035   198.609
```

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -1.451e+01  1.003e-01 -144.61  <2e-16 ***
X1           1.477e+00  6.661e-02   22.17  <2e-16 ***
X2           1.010e+00  8.043e-04 1256.22  <2e-16 ***
X3           6.889e-01  2.123e-03  324.57  <2e-16 ***
X4          -8.847e-02  2.703e-04 -327.29  <2e-16 ***
X5          -8.513e-02  6.015e-03  -14.15  <2e-16 ***
X6          -3.949e+00  5.900e-02  -66.93  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 15.51 on 327339 degrees of freedom
Multiple R-squared:  0.8792,    Adjusted R-squared:  0.8792
F-statistic: 3.97e+05 on 6 and 327339 DF,  p-value: < 2.2e-16
```

For letter b, c, d, and e

(b) Estimated Regression Equation

The estimated multiple linear regression model is:

$$\hat{Y} = -14.51 + 1.477X_1 + 1.010X_2 + 0.6889X_3 - 0.08847X_4 - 0.08513X_5 - 3.949X_6$$

(c) Interpretation of X_2

The coefficient of X_2 (departure delay) is 1.010. This means that for every additional one-minute increase in departure delay, the arrival delay is expected to increase by approximately 1.010 minutes, assuming all other variables constant.

(d) Overall Significance Test

We test the hypotheses:

$$H_0 : \beta_1 = \beta_2 = \cdots = \beta_6 = 0$$

$$H_a : \text{At least one } \beta_i \neq 0$$

From the regression output, the F-statistic is 3.97×10^5 with a p-value $< 2.2 \times 10^{-16}$.

Since the p-value is less than 0.05, we reject H_0 at the 5% significance level. Therefore, we say that at least one predictor is significantly related to arrival delay.

(e) Coefficient of Determination

The coefficient of determination is:

$$R^2 = 0.8792$$

This indicates that approximately 87.92% of the variability in arrival delay is explained by the predictors X_1, X_2, X_3, X_4, X_5 , and X_6 . The remaining 12.08% of the variability is due to other factors not included in the model or random variation.